

AYUSH PATEL

Software Engineer — AI Core Team, Samsung Research HQ · President, IIT Alumni Association of Korea (IITAAK)

B.Tech, Computer Science and Engineering
Indian Institute of Technology, Delhi

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ACADEMIC DETAILS

Year	Degree	Institute	CGPA/Percentage
2016-2020	B.Tech in Computer Science and Engineering	Indian Institute of Technology Delhi	9.040/10.0
2016	Class XII, CBSE	O. P. Jindal School, Raigarh (C.G.)	97.8%
2014	Class X, CBSE	O. P. Jindal School, Raigarh (C.G.)	10/10

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 192** in JEE Mains and **All India Rank 575** in JEE Advanced among 2 million students
- State Topper** with **97.8%** — **top 0.1%** among 1.06 million candidates in AISSCE Class XII, 2016 (CBSE)
- Selected as **KVPY Scholar** under 'Kishore Vaigyanik Protsahan Yojana' organised by Indian Institute of Science
- Awarded **IIT Delhi Merit Scholarship** for being in the **top 7%** among 850 students in the 1st and 2nd semester

CORPORATE EXPERIENCE/ INTERNSHIPS

AI Core Team <i>Software Engineer (Artificial Intelligence R&D)</i>	Samsung Research (HQ), South Korea <i>Dec 2025 - Present</i>
- Build core AI technology powering intelligent experiences across Samsung products and services worldwide - Collaborate with research labs and business units to bring AI research into Samsung's global product ecosystem	
SmartHome AI Lab <i>Software Engineer (Artificial Intelligence R&D)</i>	Samsung Research (HQ), South Korea <i>Jul 2024 - Dec 2025</i>
- Built AI experiences for millions of connected homes across the SmartThings ecosystem - Developed AI-driven home intelligence connecting devices, users and services across the smart home	
Strategic Partnership Management & Business Strategy <i>Solution Architect and Strategic Partnership/ Business Strategy</i>	Samsung Networks (HQ), South Korea <i>Nov 2023 - Jul 2024</i>
- Design and manage the System Integration (SI) Solution Architecture of 5G/O-RAN for various COTS, CaaS, RU/MMU - Layout process for System Integration and service delivery of Service Management and Orchestration (SMO) and rApps - Strategic Partnership Management (new business strategies, maintain relation, etc.) for Hyperscalers and COTS vendors	
Advanced Cloud Native Solutions <i>Software R&D (Cloud Platform/Infrastructure Development & Operations)</i>	Samsung Networks (HQ), South Korea <i>Oct 2020 - Oct 2023</i>
- Develop Cloud Native Load Balancer for various protocols for 5G vCore & vRAN using Kubernetes Operator Framework - Work with cloud native tools & plugins like helm, ansible, multus, cilium, DPDK, SRIOV etc. on public & private cloud - Contribute to open-source projects like whereabouts, multus and other libraries and participate in community discussions	
Tech Lead <i>Co-founder (Management & Operations)</i>	Halanx Technologies Pvt. Ltd. <i>Jan 2020 - May 2020</i>
- Led the Technical team comprising of App Developers, Backend Developers, Frontend Developers and UI/UX Designers - Developed Rental Housing Solution to reduce the operating cost of owners and provide affordable housing to the tenants	
Cloud Network Platform Development <i>Summer Internship (Cloud Software R&D)</i>	Samsung Electronics (HQ), South Korea <i>May 2019 - July 2019</i>
- Developed a Generic Toolkit for Profiling, Debugging, Automation and Monitoring Health of any Kubernetes Cluster - Improved the available Samsung Cloud Platform (SCP) Toolkits for incorporating various CNIs (eg. OpenStack Kuryr)	
DashCam Car Accident Prediction <i>Summer Internship (AI/ML Research Project)</i>	ITRI, Hsinchu, Taiwan <i>May 2018 - July 2018</i>
Worked with OpenCV, scikit-learn, YOLO, Tensorflow etc. to train car accidents prediction model for self-driving cars	

LICENSES & CERTIFICATIONS

- **Certified Kubernetes Administrator (CKA)** and **Linux Foundation Certified System Administrator (LFCS)**
- **Cloud Specialist** — Samsung Electronics internal certification for cloud platform & infrastructure specialists
- **TOPIK Level 5** (advanced Korean proficiency) · **KIIP Level 6** · **KINAT** — Korean language & integration

LEADERSHIP & COMMUNITY

- **President of IIT Alumni Association of Korea** (IITAAK: iitalumkorea.org) (2023 - Present); **Vice President, Public Relations** (2021 - 2023)
- **Speaker** at the **India–Korea Forum on AI 2026**, Ewha Womans University, Seoul (July 2026)
- **Host** of the Fun Cinema & VFX Booth at **India Day Seoul 2026**, Embassy of India (May 2026)
- **Executive Secretary, Indian Students and Researchers in Korea (ISRK)**; **Core Team Member, Indians in Korea (IIK)** (2025 - Present)
- **Microsoft Student Partner India**, Delhi-NCR Region (2017 - 2019)
- **Overall Coordinator** of Nokia Data Science Hackathon; **Executive & Developer**, DevClub IITD (2017 - 2018)
- **Technical Secretary cum Coordinator** — BHM, BSA, BRCA, AAIP, SAC and **Tryst**, IIT Delhi (2017 - 2019)

MAJOR PROJECTS

- Automotive Security System** B.Tech Project, IIT Delhi (2019 - 2020)
- Studied the architecture of Controller Area Network (CAN) bus and Plug-in Hybrid Electric Vehicle (PHEV) Technology
 - Devised an improved architecture which tackles the existing security vulnerability without hampering the performance
- Nokia ML Hackathon Platform** Industry Project, IIT Delhi (2018)
- Developed a platform to host large scale Data Science Hackathons using Hadoop, Docker, JupyterHub, NodeJS, Django
- Tank Assault** DevClub IIT Delhi (2017)
- Developed a web based multiplayer game with Node.JS as Backend server and used sockets for real time communication

COURSE PROJECTS

- Baadal (IITD Computing Cloud)** Cloud Computing (September - November 2020)
- Implemented Load Balancing Module, On-Demand VM migration system and Resource Monitoring dashboard in Baadal
- Extended Xv6 Operating System** Operating Systems (March - May 2019)
- Implemented various system calls, interrupt handlers, copy-on-write (COW) fork and deployed a fair scheduling algorithm
- Multicycle ARM Processor** Computer Architecture (March - April 2018)
- Developed a Multicycle processor for the ARM ISA in VHDL on FPGA, implementing AHB Lite Bus for connections
- Prolog Interpreter** Programming Languages (March - April 2018)
- Implemented an interpreter for Prolog in OCaml using OCaml-lex for token generation and OCaml-yacc for parsing

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Java, C#, Python, JavaScript, PHP, SQL, Bash, VHDL, LaTeX, Ocaml, Prolog, Scala, XML, Go, Rust
- **Softwares and Frameworks:** Kubernetes, Docker, Helm, Ansible, Multus, Cilium, DPDK, SR-IOV, OpenStack, OpenCV, OpenMP, MPI, CUDA, Django, NodeJS, Angular, React, Android Studio, React-Native, Flutter, Git, Apache Hadoop, Spark, PyTorch, Tensorflow, KVM, QEMU, Unity, MATLAB
- **Languages Spoken:** English, Hindi, Korean (TOPIK Level 5)